

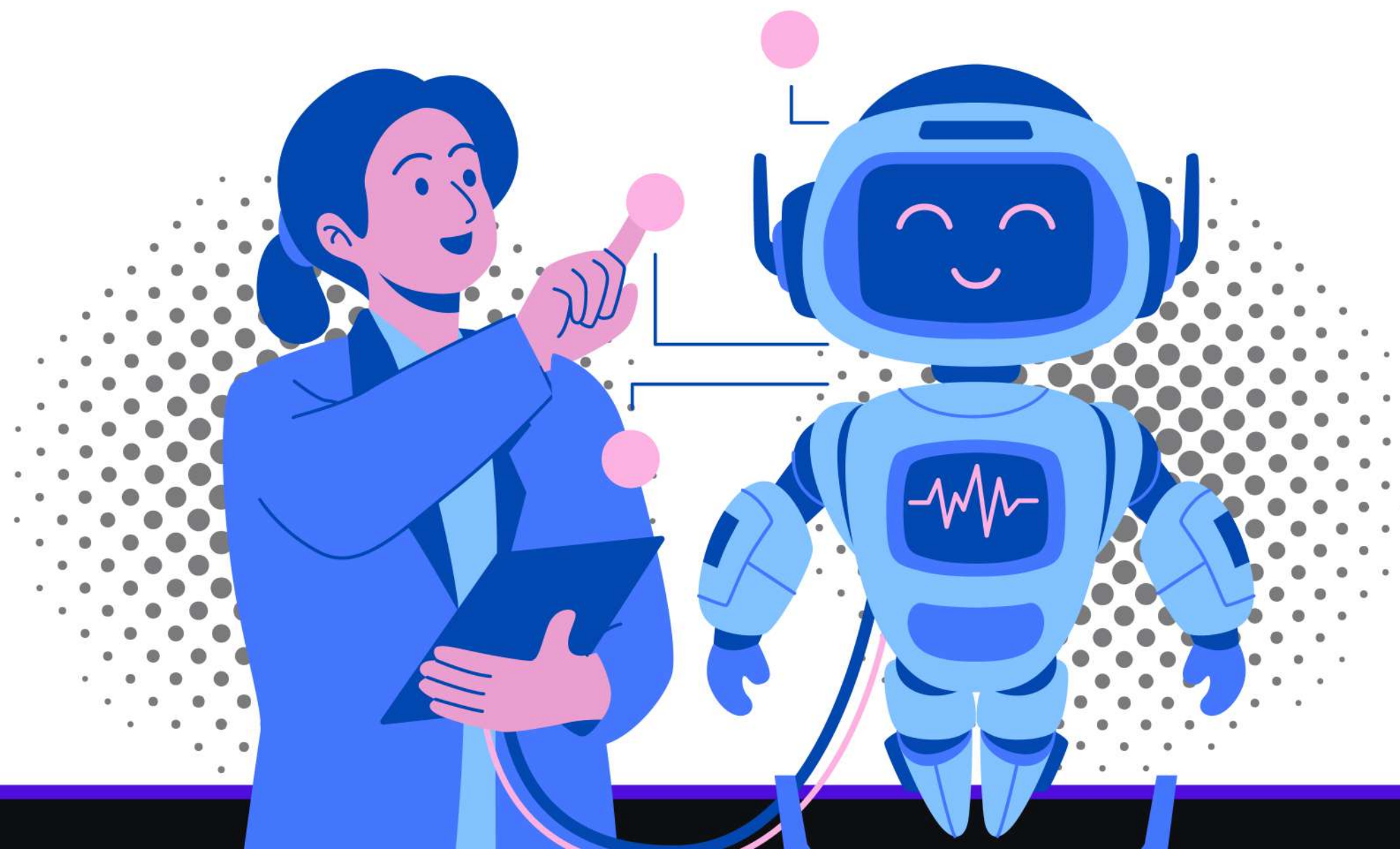


by Kunal Das



# BEYOND THE HYPE

BUILDING A SUSTAINABLE, PROFITABLE AI  
STRATEGY

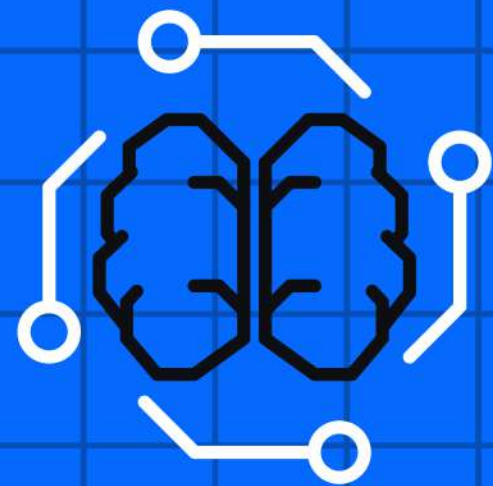


DATE December 21, 2025

LOCATION Bangalore



Hello



# ***KUNAL DAS***

Developer Advocate APAC, CAST AI

Organizer of  
CNCF Kolkata,  
Cloud Computing Circle,  
Hashicorp User Group Bangalore

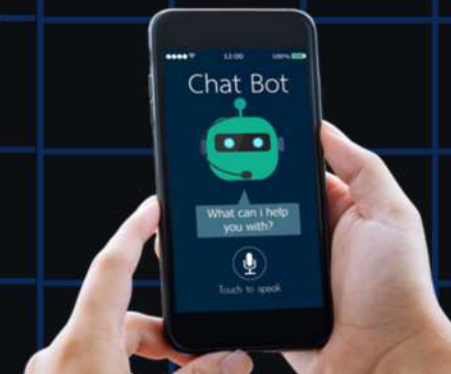
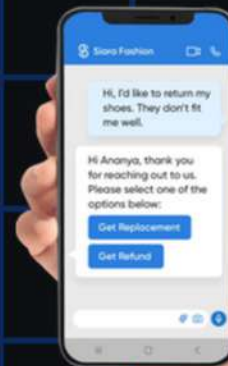
7x Azure, 1x Hashicorp Certified, FinOps  
Certified Engineer

# How we use LLMs

End users



UI/UX



API



# LET'S CHECK SOME PROMPTS

...

## PROMPT1

*What is Kubernetes and what are its main components?*



GPT-3.5-turbo



Claude 3 Haiku



Llama 3.1:8b

## PROMPT2

*Analyze this Kubernetes cluster's resource utilization data and provide 3 key optimization recommendations:*

- 40% CPU usage across 50 nodes
- 60% memory utilization
- 15 pods in pending state
- Average pod startup time: 45 seconds



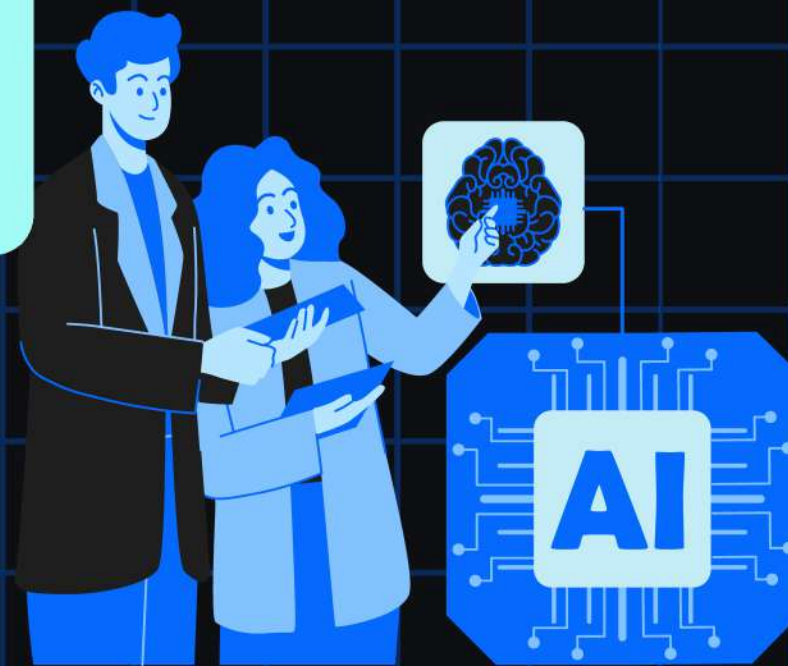
GPT-4o-mini



Claude 3 Sonnet



Llama 3.1:70b



# LET'S CHECK SOME PROMPTS



## PROMPT 3

*Design a multi-region Kubernetes disaster recovery strategy for a financial services application that:*

- *Must maintain 99.99% uptime*
- *Handles sensitive PII data*
- *Requires <100ms latency for transactions*
- *Operates across AWS and GCP*
- *Has compliance requirements for GDPR and SOC2*

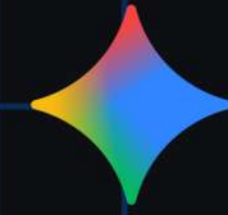
*Provide architecture decisions, security considerations, cost implications, and implementation phases.*



GPT-4o



Claude 3.5 Sonnet



Gemini 1.5 Pro

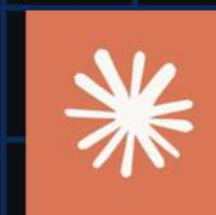
## PROMPT 4

*Create a Python script that:*

- *Connects to a Kubernetes cluster using the official client*
  - *Identifies pods with CPU usage > 80%*
  - *Automatically scales their deployments*
  - *Sends Slack notifications with scaling actions*
  - *Includes error handling and logging*
- 6. Add unit tests for the scaling logic*



GPT-5



Claude 4 Opus



Deepseek-R1:14b



# PURPOSE DRIVEN LLM

## Search & Fact checking

These models excel at grounding responses in current data and cross-referencing multiple sources for truth verification.



## Deep research thinking

These models handle intricate queries that demand deep understanding, context retention, and nuanced decision-making.



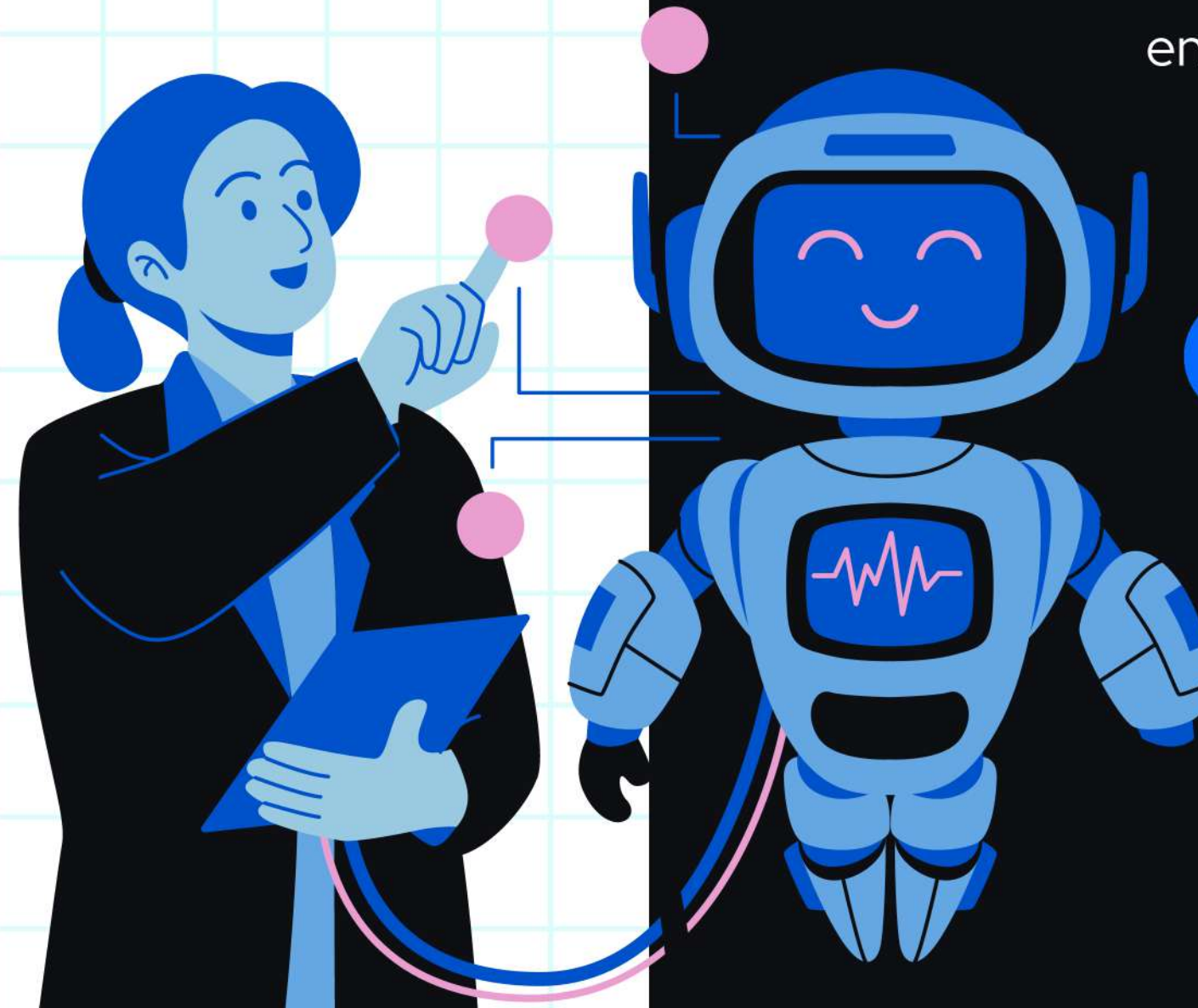
## Integrations

These lightweight models provide compatible endpoints across all providers, ensuring smooth workflow automation.

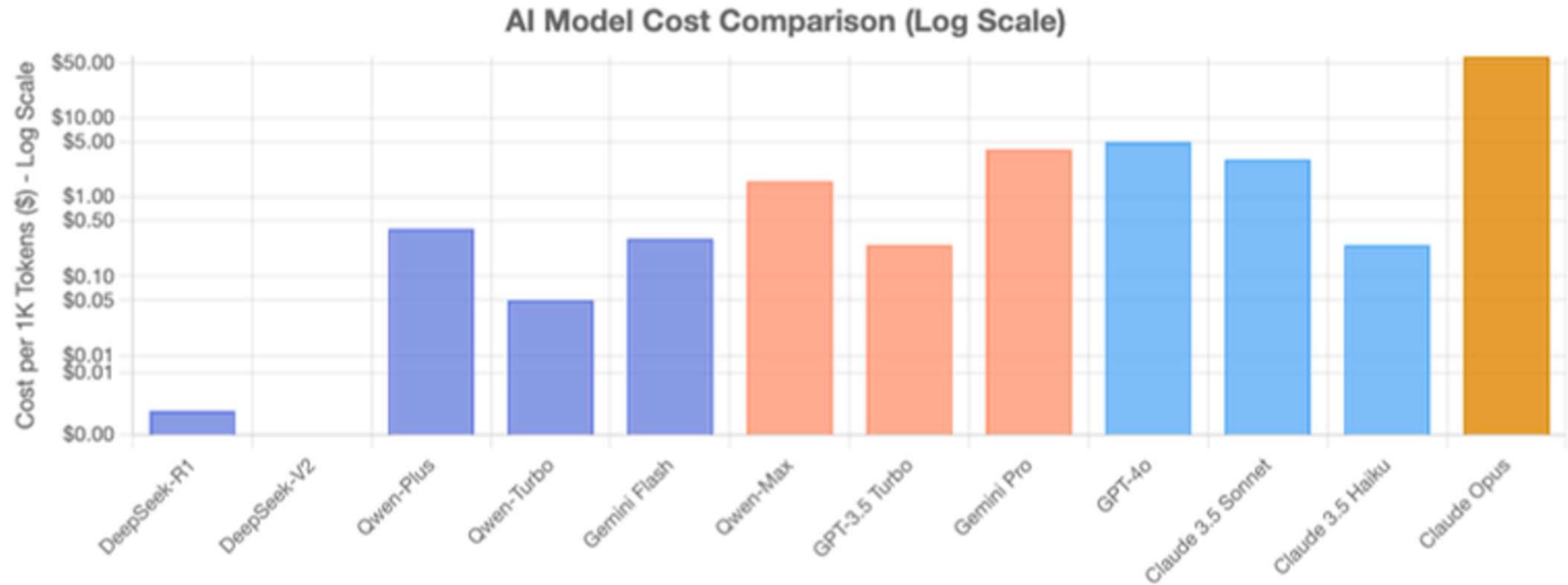
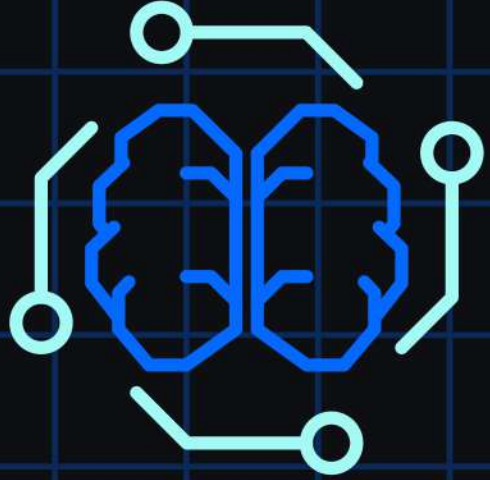


## Coding / Development

These models understand code context, suggest optimizations, and can handle complex refactoring operations.



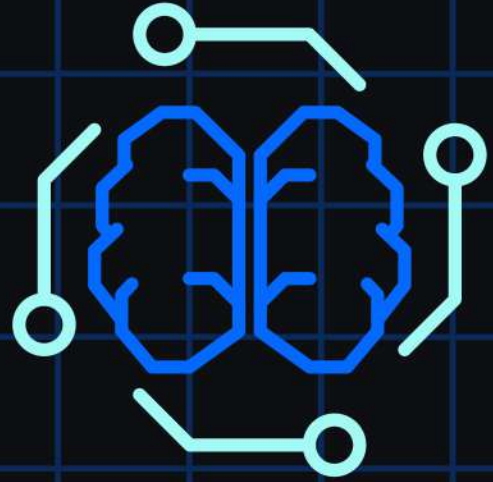
# Let's Talk About Money



⚠️ **CRITICAL INSIGHT:** Using Claude 3 Opus (\$60/1K) vs DeepSeek-V2 (\$0.001/1K) = 60,000x cost difference!



# Let's Talk About Money



## 💎 Budget Champions

< \$0.50/1K tokens  
DeepSeek, Gemini Flash,  
Qwen, Alibaba Models

## ⚡ Balanced Performance

\$0.50 - \$5.00/1K  
tokens  
GPT-4o, Gemini Pro,  
OpenAI GPT-3.5

## 🎯 Premium Quality

\$5.00 - \$15.00/1K  
tokens  
Claude 3.5 Sonnet, GPT-4o Mini

## 🏆 Enterprise Grade

> \$15.00/1K tokens  
Claude 3 Opus, Claude 3.5  
Haiku



# LLM HOSTING

## Self-Hosting in VM

- Deploy models directly on virtual machines
- Best for: Low-volume workloads, testing, single-application use cases

## Kubernetes Deployment

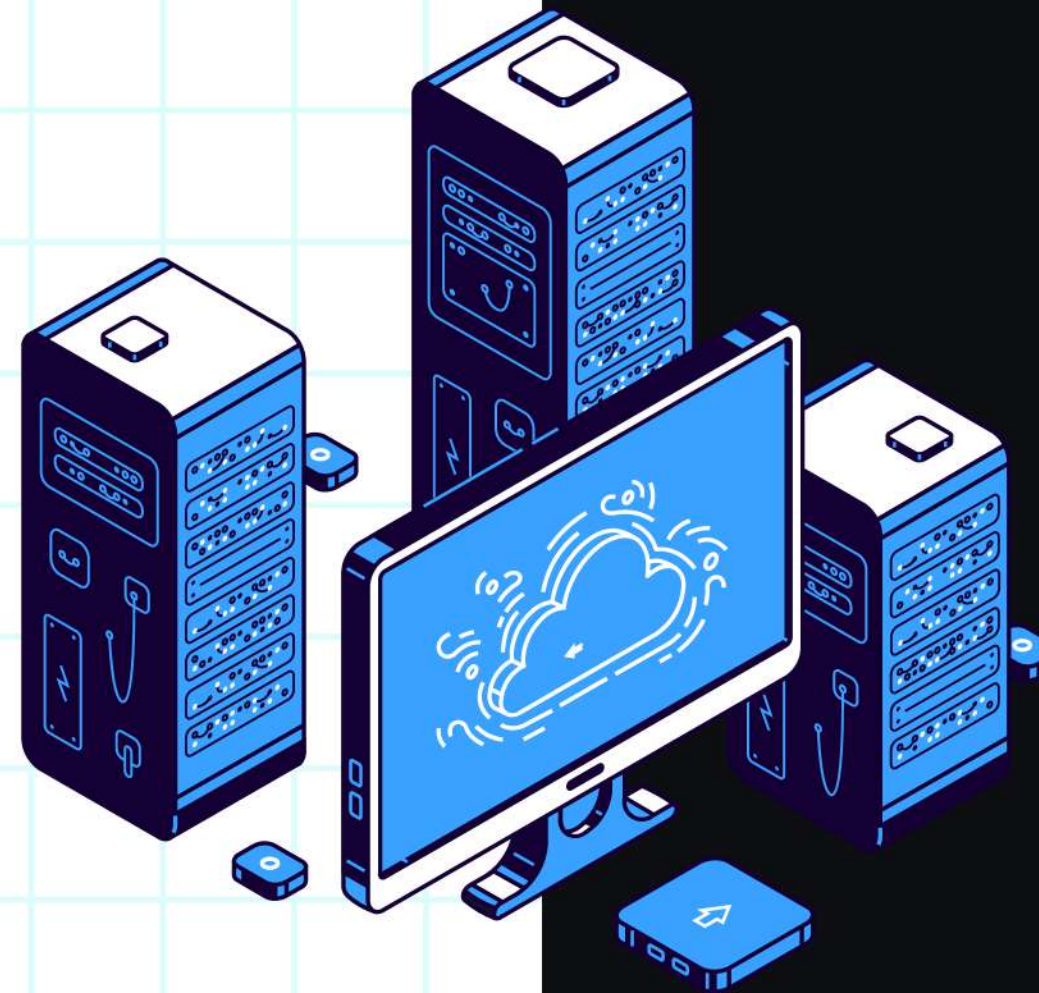
- Auto-scaling LLMs with in your cluster
- Best for: High availability, multi-tenant apps, microservices architecture

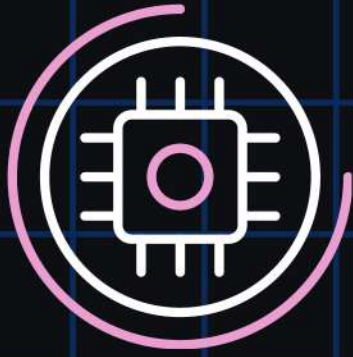
## API-Based

- write logic to call different APIs as per requirements
- Best for: Less setup time, rapid iteration

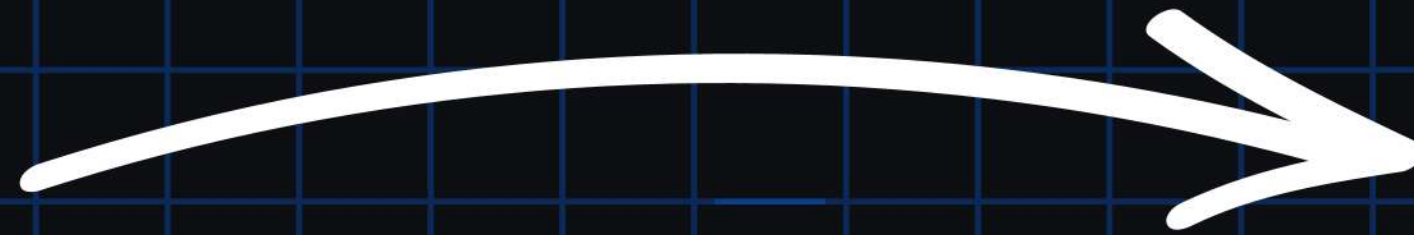
## Fallback Mechanism

- Automatic failover when primary models fail or rate-limit
- Best for: Production systems, SLA requirements, business continuity





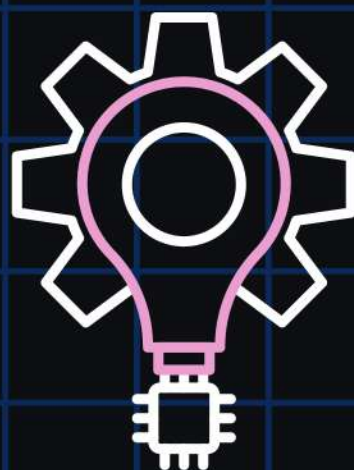
# AI ENABLER

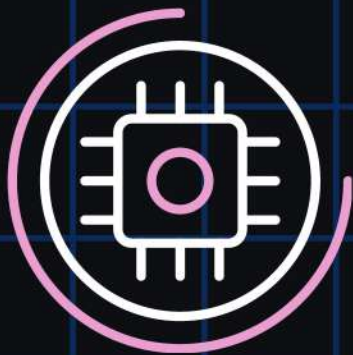


Open AI models

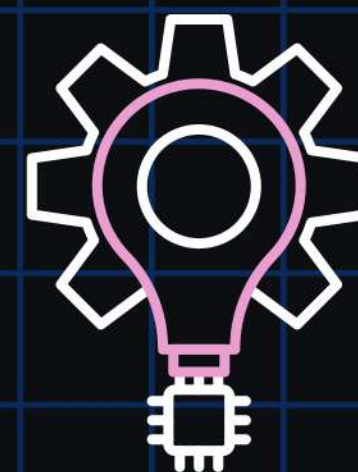
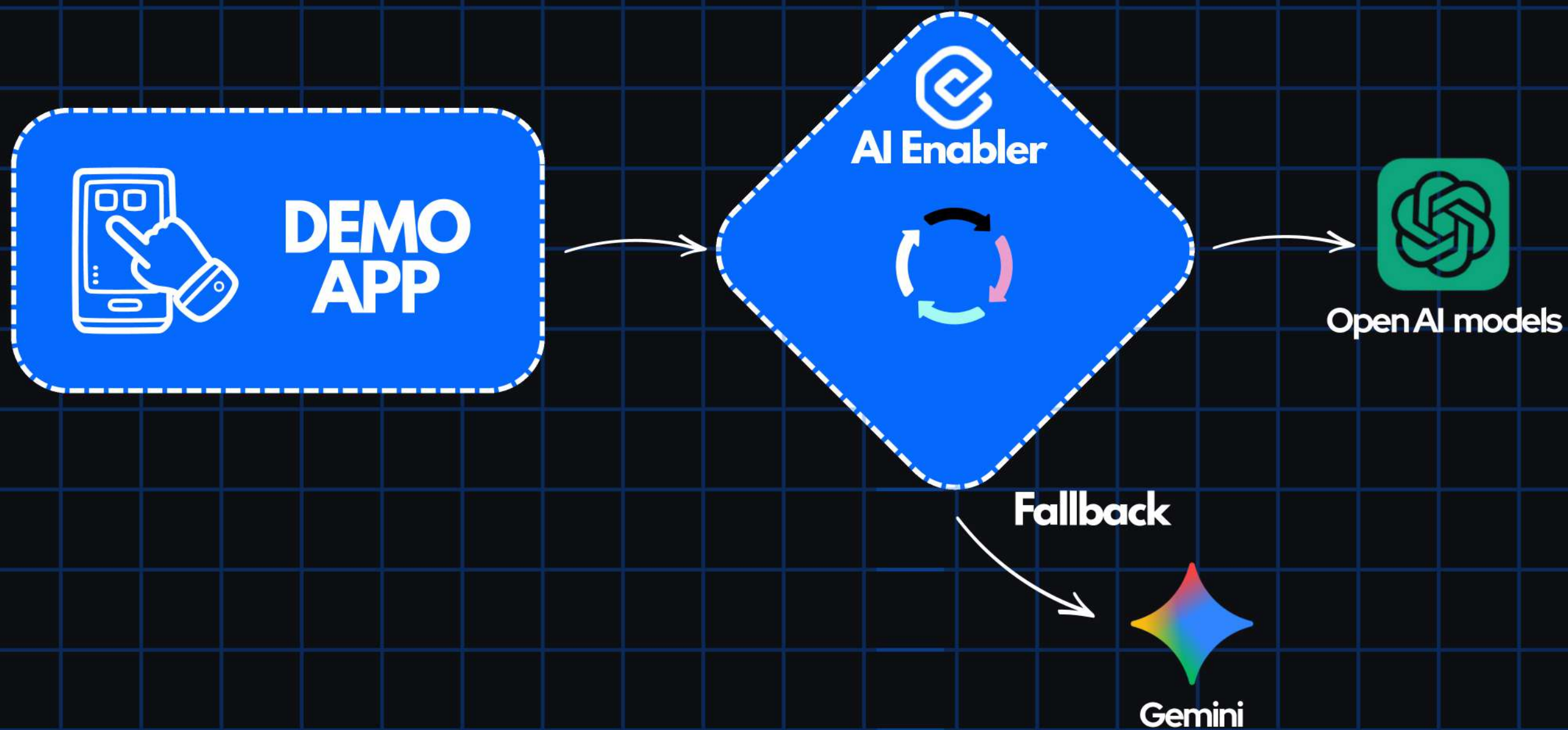


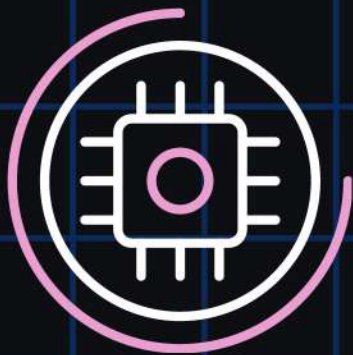
Open AI models



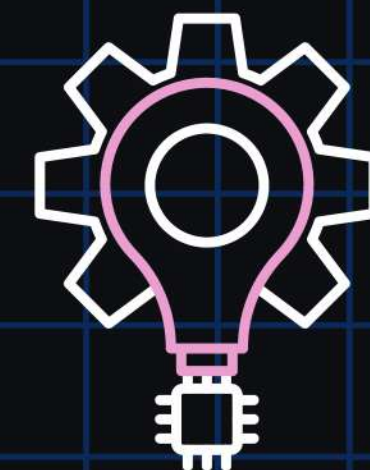
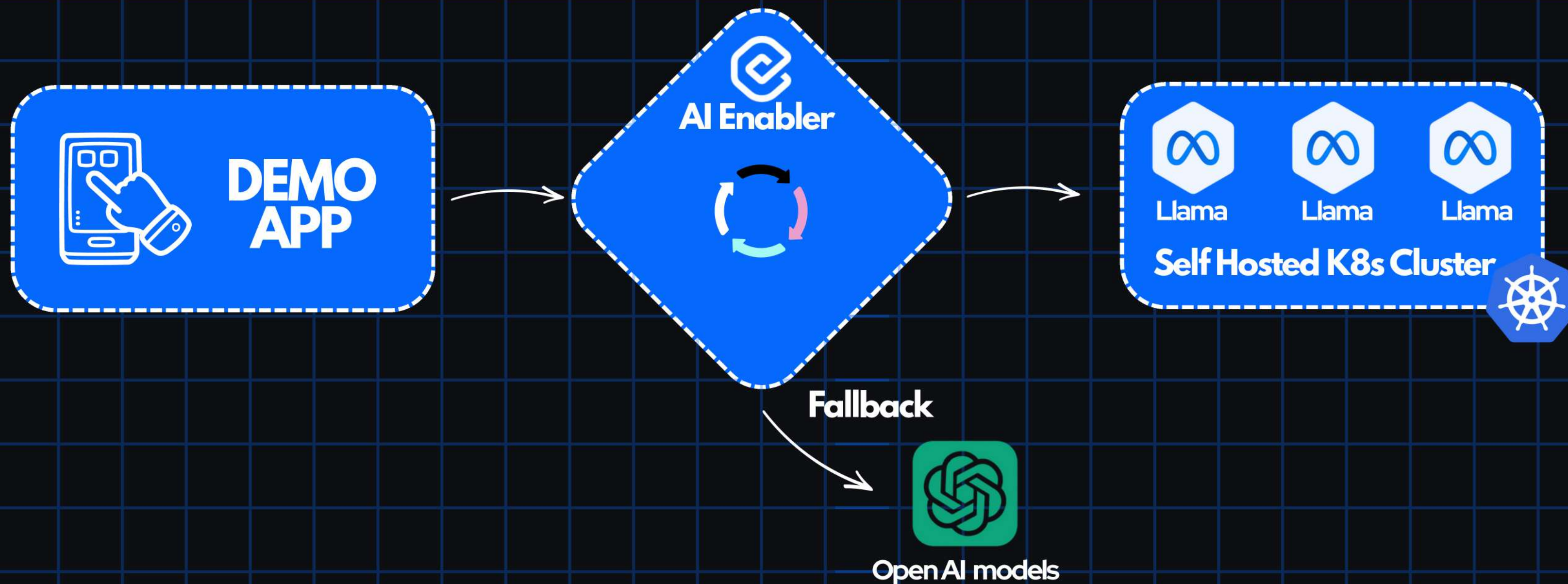


# AI ENABLER





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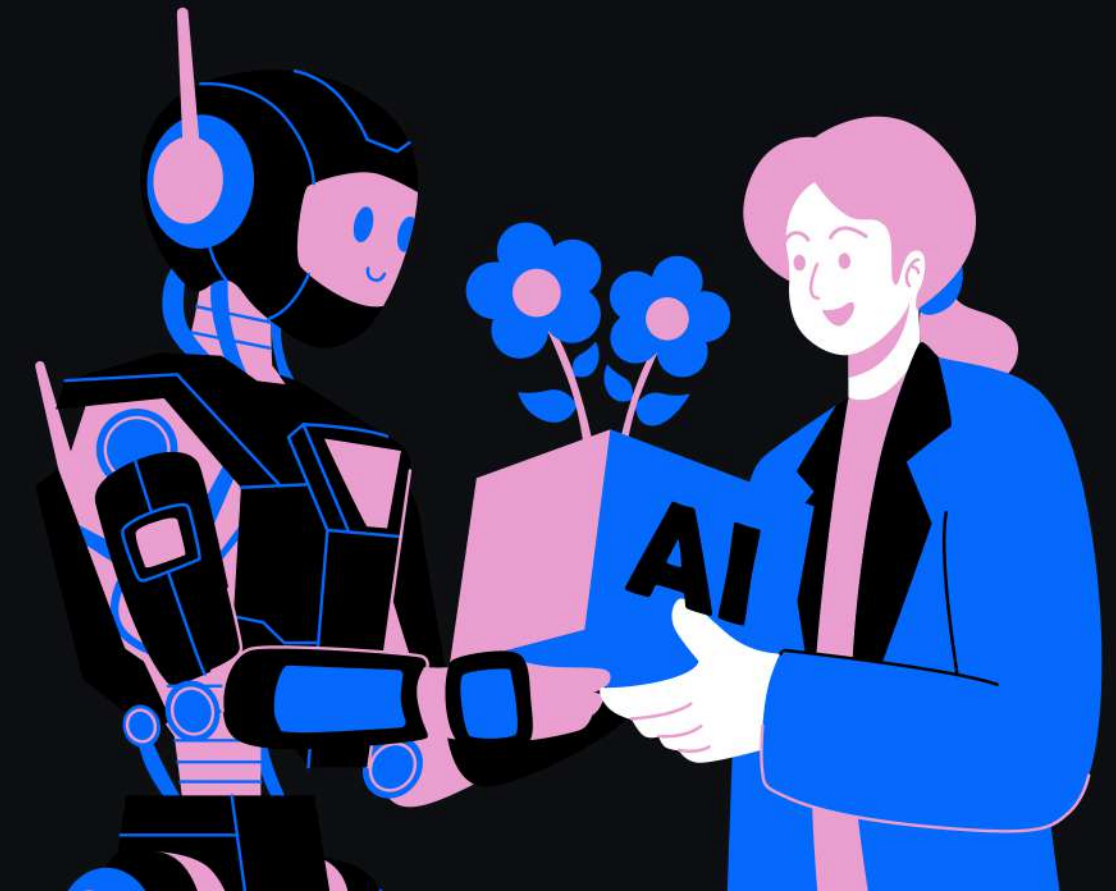


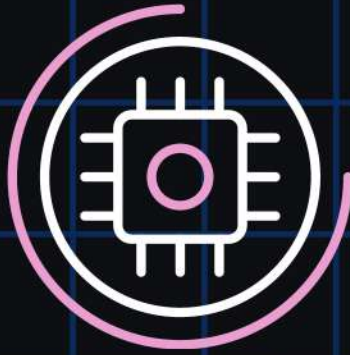
# WHAT WE ACHIEVED

**Multi-Provider Freedom**

**Enterprise-Ready Reliability**

**Cost Savings**





# RELAIBILITY??

**Cybernews**

**Thousands of users report Amazon issues**

1 day ago



**Reuters**

**Amazon says AWS cloud service back to normal after outage disrupts businesses worldwide**

Disruptions to its servers can cause outages across websites and platforms - ranging from food delivery apps to gaming platforms and airline...

2 weeks ago



**CRN Magazine**

**Outflare Blames Google Cloud For Mass Services Outages**

Outflare blamed an outage at Google Cloud for a number of products and services by content delivery network vendor Cloudflare Thursday.

Jun **Mint**

1 hour ago



**Zoom Bangla News**

**Microsoft Outage Today: Why Are Azure, 365, and Teams Down Across the U.S.?**



**Times Now**

**Microsoft Azure Down: Server Outage Impacts Multiple Services Including 365,...**

6 minutes ago



**ET The Economic Times**

**Azure down: Thousands of users complain about outage; here's Microsoft's latest...**

2 hours ago



**Mashable**

**Google Cloud down: The GCP outages explained**

Google reported that Google Cloud Platform (GCP), its popular cloud hosting service, experienced a problem Thursday afternoon.

12 Jun 2025



**WIRED**

**Security News This Week: Amazon Explains How Its AWS Outage Took Down the Web**

AWS confirmed in a "post-event summary" on Thursday that its major outage on Monday was caused by Domain System Registry failures in its...

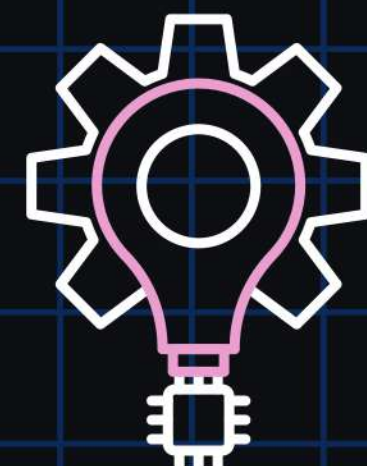
2 weeks ago

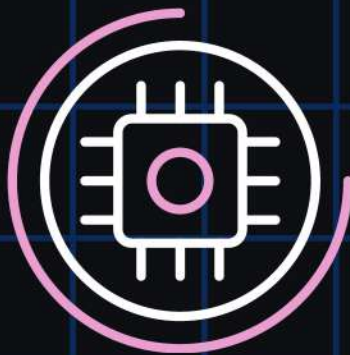


**amazon**

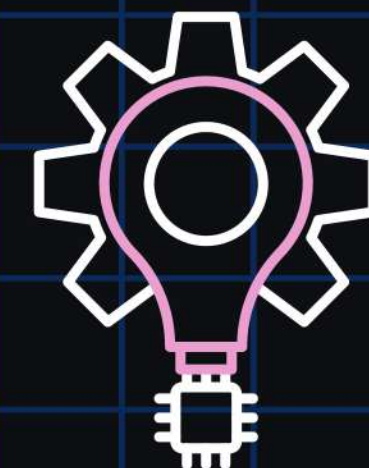
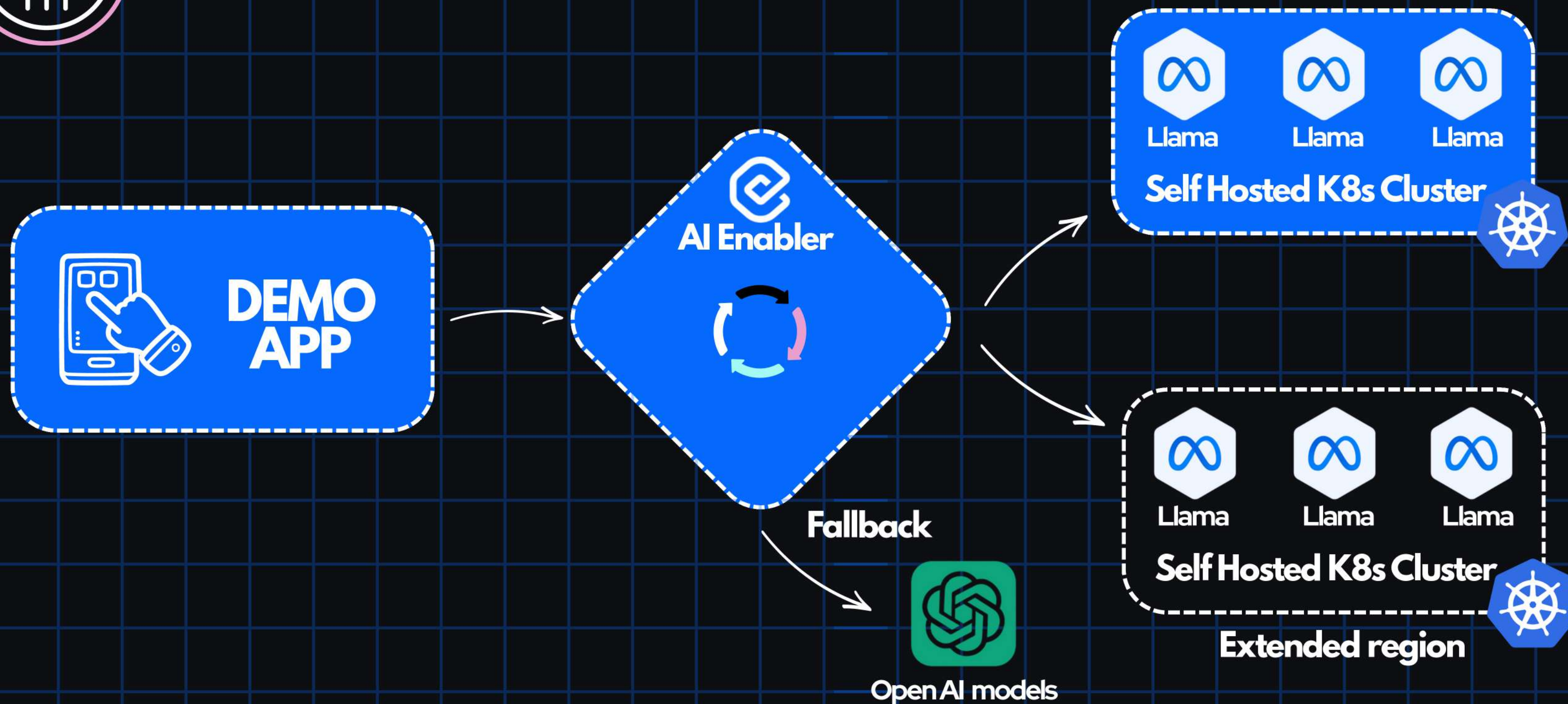
**BREAKING NEWS**

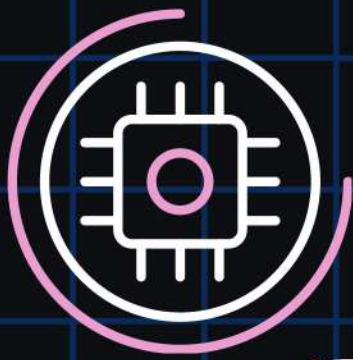
**AMAZON'S CLOUD SERVICE IS DOWN AGAIN**



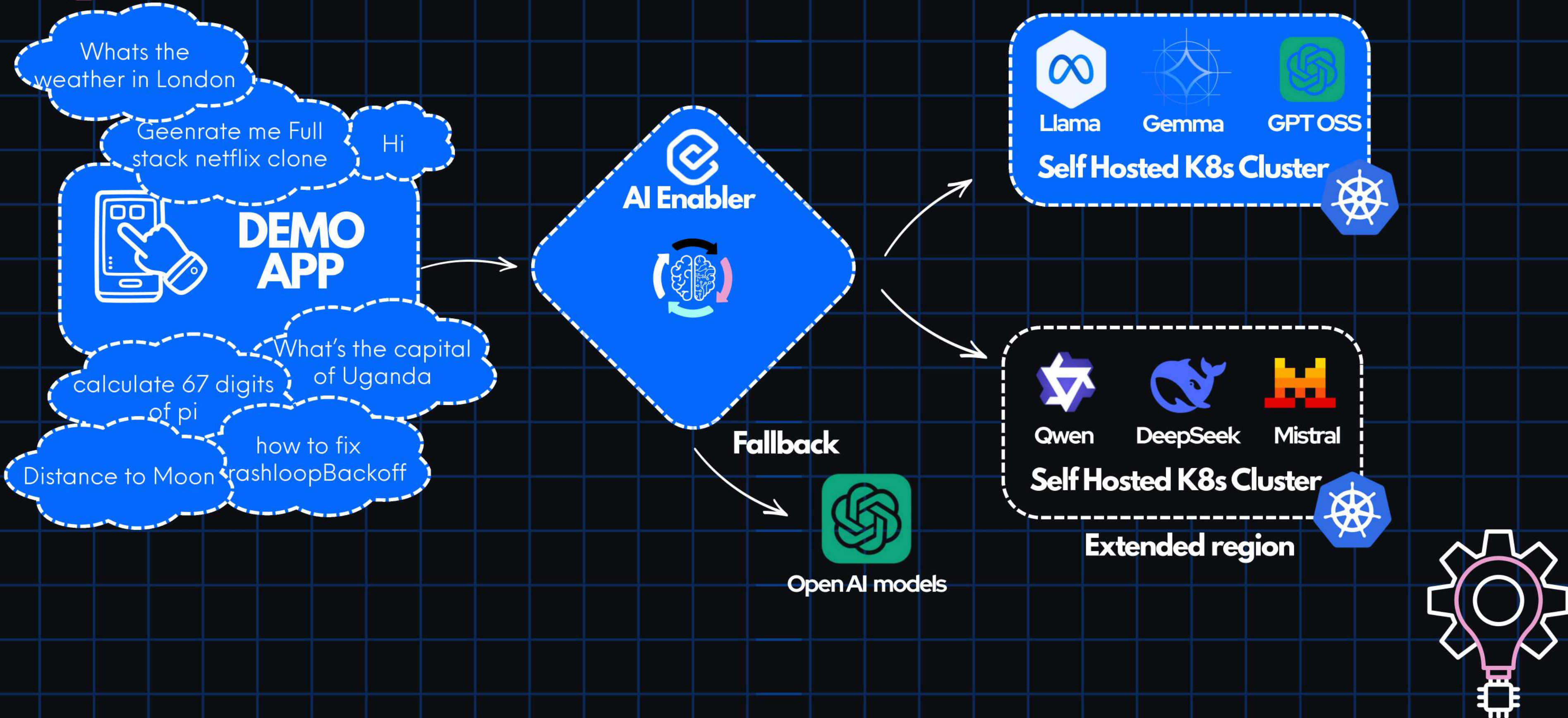


# AI ENABLER





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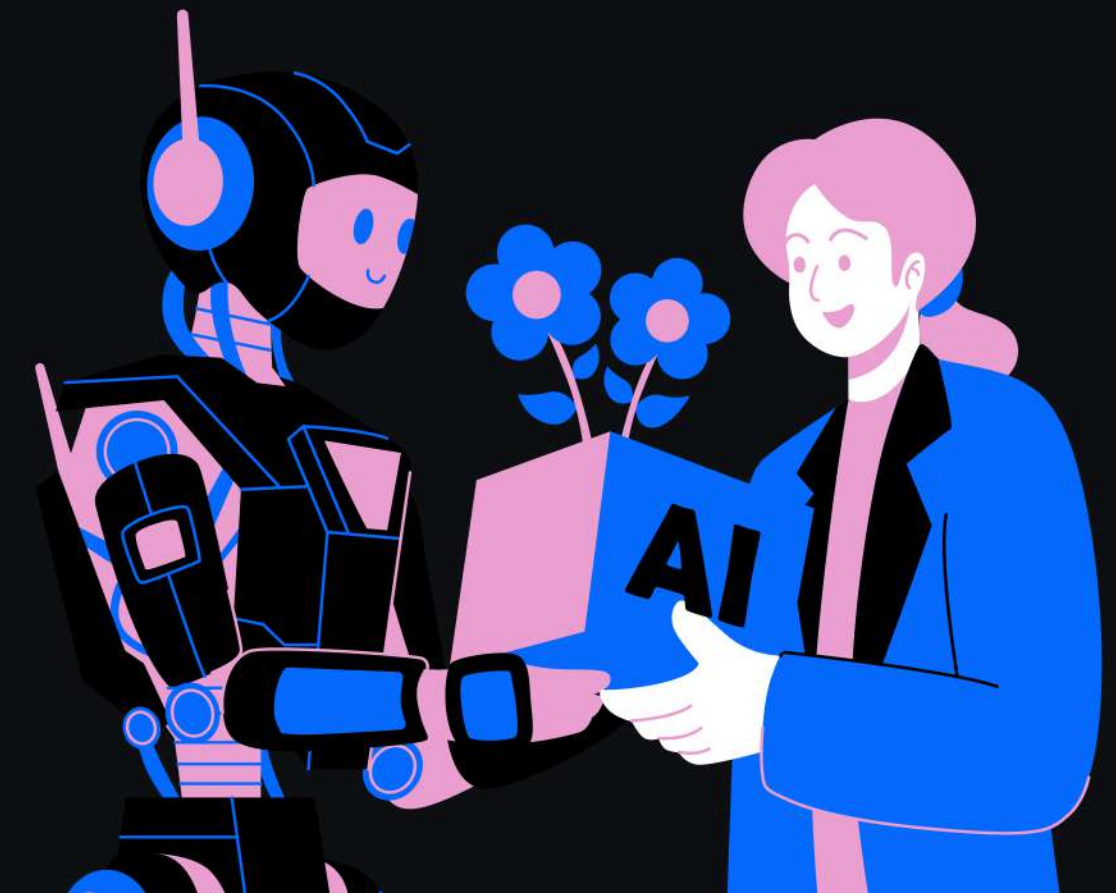
# WHAT WE ACHIEVED

**Multi-Provider Freedom**

**Enterprise-Ready Reliability**

**Intelligent Cost Savings**

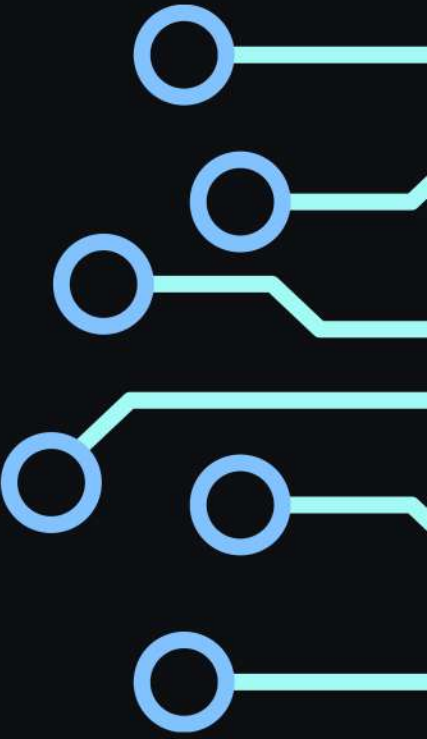
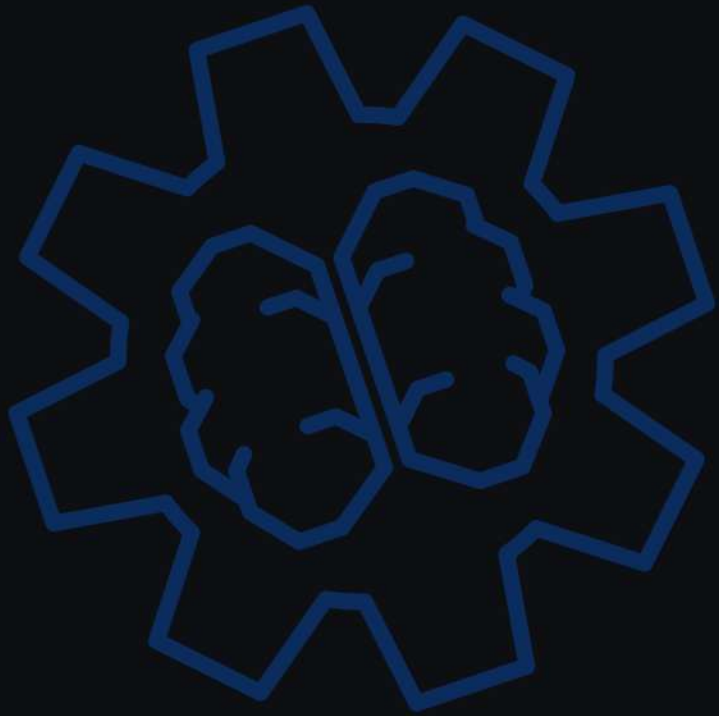
**Quality-First Routing**



# CONCLUSION

- ✓ Intelligent routing saves money
- ✓ Quality doesn't have to cost premium prices
- ✓ Open-source Llama competes with proprietary models
- ✓ CAST AI makes it simple to implement

**“The best LLM isn't the most expensive one - it's the RIGHT one for each task. ”**



# THANK YOU

For your attention



PING ME  
ANYTIME TO  
KNOW MORE

